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Kelas : TI – 25 – PA

TUGAS 9

```
1  #include <iostream>
2  #define MAX 5
3
4  using namespace std;
5
6  class Stack {
7  private:
8      int top;
9      int arr[MAX];
10
11 public:
12
13     Stack() {
14         top = -1;
15     }
16
17     bool isFull() {
18         return top == MAX - 1;
19     }
20
21     void push(int data) {
22         if (isFull()) {
23             cout << "Stack Penuh!" << endl;
24             return;
25         }
26         arr[++top] = data;
27     }
28
29     void display() {
30         cout << "Output :" << endl;
31         for (int i = top; i >= 0; i--) {
32             cout << "| " << arr[i] << " |" << endl;
33             cout << "-----" << endl;
34         }
35     }
36 };
37
38 int main() {
39     Stack s;
40
41     s.push(621);
42     s.push(999);
43     s.push(201);
44     s.push(94);
45     s.push(124);
46
47     s.display();
48
49     return 0;
50 }
```

```
Output :
| 124 |
-----
| 94  |
-----
| 201 |
-----
| 999 |
-----
| 621 |
-----

-----
Process exited after 0.29 seconds with return value 0
Press any key to continue . . . |
```

```

1  #include <iostream>
2  using namespace std;
3
4  struct Node {
5      int data;
6      Node* next;
7      Node* prev;
8  };
9
10 class DoublyLinkedList {
11 private:
12     Node* head;
13     Node* tail;
14
15 public:
16
17     DoublyLinkedList() {
18         head = NULL;
19         tail = NULL;
20     }
21
22     void append(int data) {
23         Node* newNode = new Node();
24         newNode->data = data;
25         newNode->next = NULL;
26         newNode->prev = NULL;
27
28         if (head == NULL) {
29             head = newNode;
30             tail = newNode;
31         } else {
32             tail->next = newNode;
33             newNode->prev = tail;
34             tail = newNode;
35         }
36     }
37
38     void display() {
39         Node* temp = head;
40         cout << "Bentuk Doubly Linked List:" << endl;
41         cout << "NULL <- ";
42         while (temp != NULL) {
43             cout << "[ " << temp->data << " ]";
44             if (temp->next != NULL) {
45                 cout << " <-> ";
46             }
47             temp = temp->next;
48         }
49         cout << " -> NULL" << endl;
50     }
51 };
52
53 int main() {
54     DoublyLinkedList dll;
55
56     dll.append(24);
57     dll.append(10);
58     dll.append(33);
59     dll.append(33);
60
61     dll.display();
62
63     return 0;
64 }

```

```
Bentuk Doubly Linked List:  
NULL <- [ 24 ] <-> [ 10 ] <-> [ 33 ] <-> [ 33 ] -> NULL  
-----  
Process exited after 0.2568 seconds with return value 0  
Press any key to continue . . .|
```